

Digital Design SET 111

Lecture (02)- Introductory Concepts

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- ☐ **Digital and Analog Quantities**
- ☐ **Binary Digits, Logic Levels, and Digital Waveforms**
- ☐ **Basic Logic Functions**
- ☐ **Programmable Logic**
- ☐ **Fixed-Function Logic Device**



☐ Digital and Analog Quantities

☐ Binary Digits, Logic Levels, and Digital Waveforms

☐ Basic Logic Functions

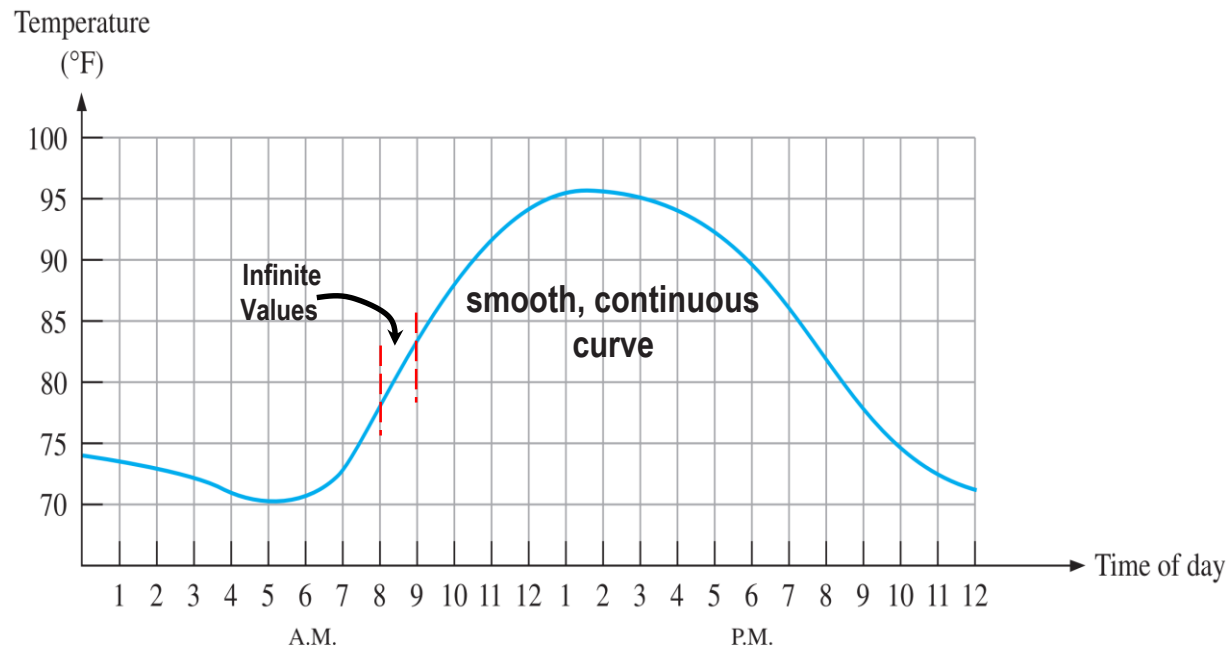
☐ Programmable Logic

☐ Fixed-Function Logic Device

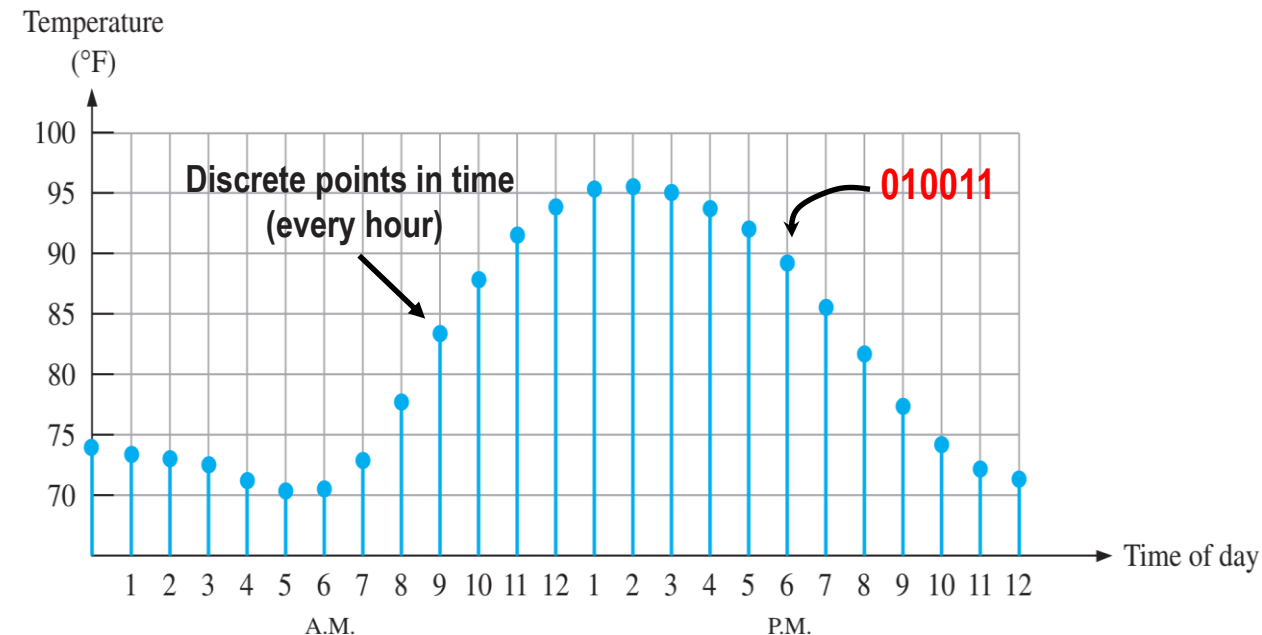
Digital and Analog Quantities (1/3)



- ❑ Electronic circuits can be divided into **two** broad categories:
 - ❑ **Analog electronics** involves quantities with **continuous** values.
 - ❑ **Digital electronics** involves quantities with **discrete** values.

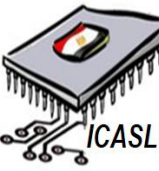


- ❑ Most things that can be measured quantitatively occur in **nature** in **analog form**



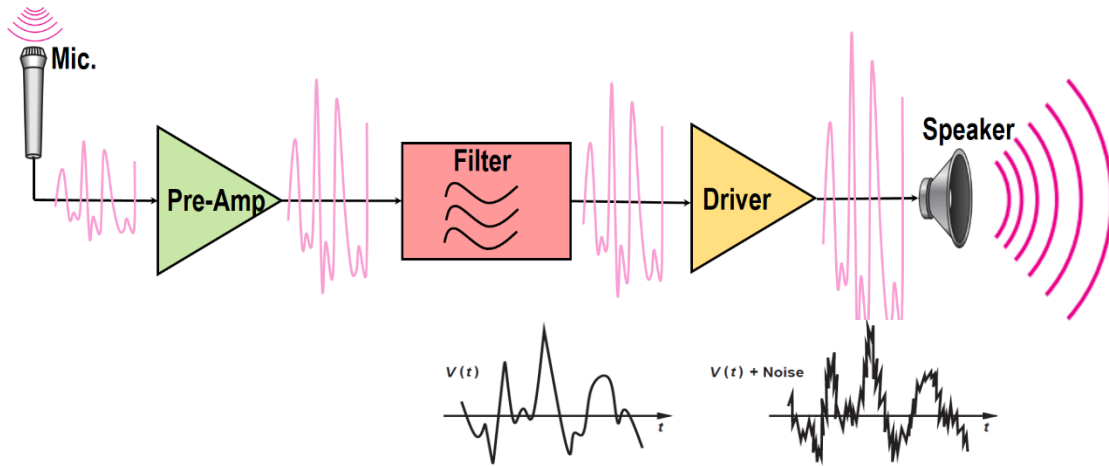
- ❑ Sampled-value representation (**quantization**) of the **analog quantity**
- ❑ Each value represented by a **dot** can be digitized by representing it as a **digital code** (series of 1s and 0s).

Digital and Analog Quantities (2/3)



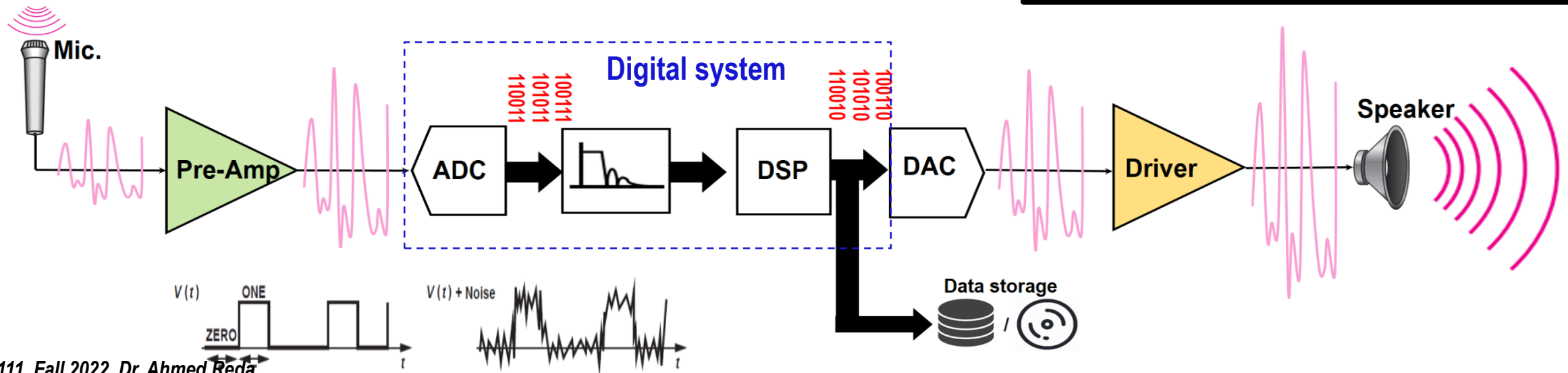
□ Analog system

Sound/Speech



□ Analog and digital system (mixed-mode system)

Sound/Speech



Digital System Advantages

High immunity to noise (Robust).
Stored in a digital memories.
Easily compressed and encrypted.



Digital System Disadvantages

Sampling may cause loss of information.
A/D and D/A consume high power.
Processing is more complex.



Digital and Analog Quantities (3/3)



Mechatronics

Power Electronics

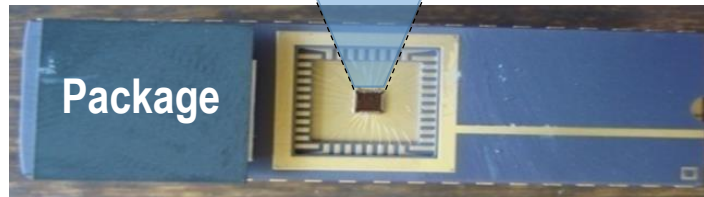
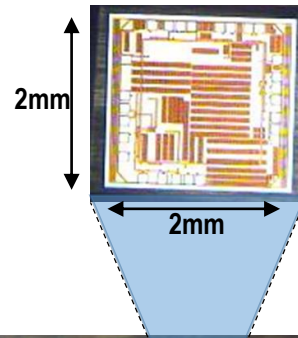
Biomedical Applications



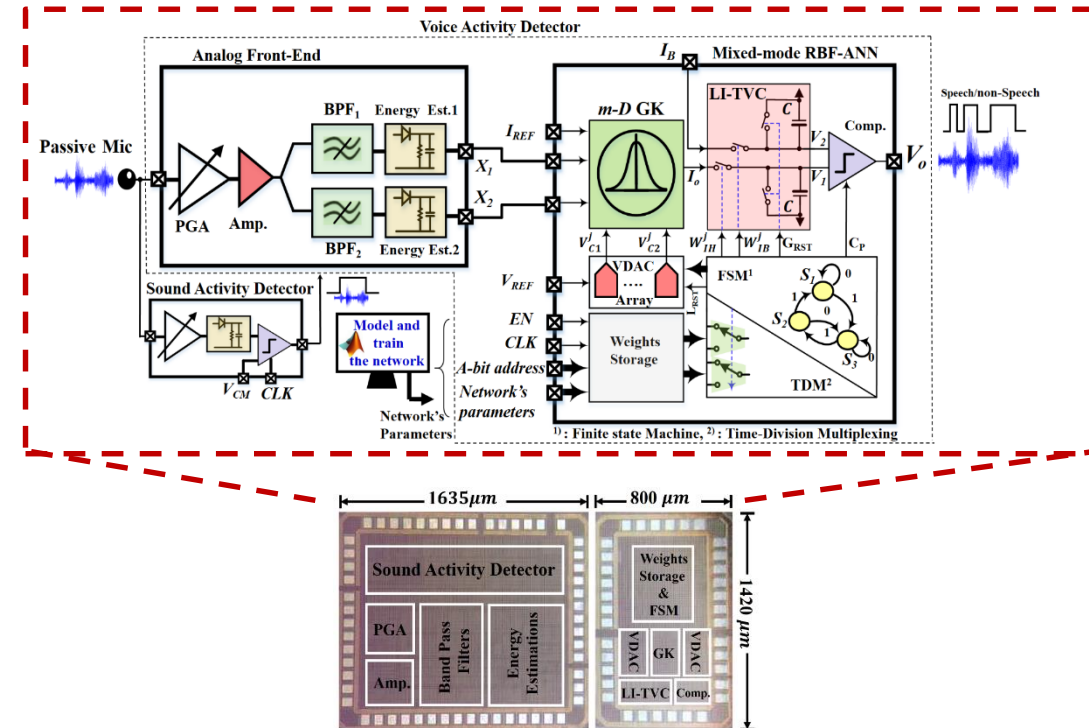
Robotic arm



Automotive assembly line



A. Reda



A. Reda



- ☐ Digital and Analog Quantities
- ☒ Binary Digits, Logic Levels, and Digital Waveforms
- ☐ Basic Logic Functions
- ☐ Programmable Logic
- ☐ Fixed-Function Logic Devices

Binary Digits, Logic Levels, and Digital waveforms (1/5)



❑ Digital electronics involves circuits and systems use only **two** possible states:

HIGH → 1 and **LOW** → 0

❑ The **two-state** number system is called **binary**.

❑ Combinations of the **two states** is called **codes** to represent, numbers, symbols, Alphabetic characters, and Instructions .

101101010 Code
↑
Bit

❑ Binary Digits

❑ Each of the two digits in the **binary** system, 1 and 0, is called a **bit**, which is a contraction of the words **binary digit**.

❑ In the **positive logic**, 1 is represented by a **HIGH** and a 0 is represented by a **LOW**

❑ In the **negative logic**, 1 is represented by a **LOW** and a 0 is represented by a **HIGH**

Binary Digits, Logic Levels, and Digital waveforms (2/5)



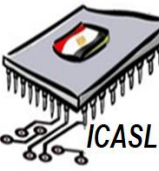
□ Logic Levels

□ The **voltages** used to represent a **High**→1 and a **Low**→0 are called **logic levels**.

□ Ideally, **High** level is represented by a **single voltage value**. Also, **Low** level is represented by another voltage value.



Binary Digits, Logic Levels, and Digital waveforms (2/5)

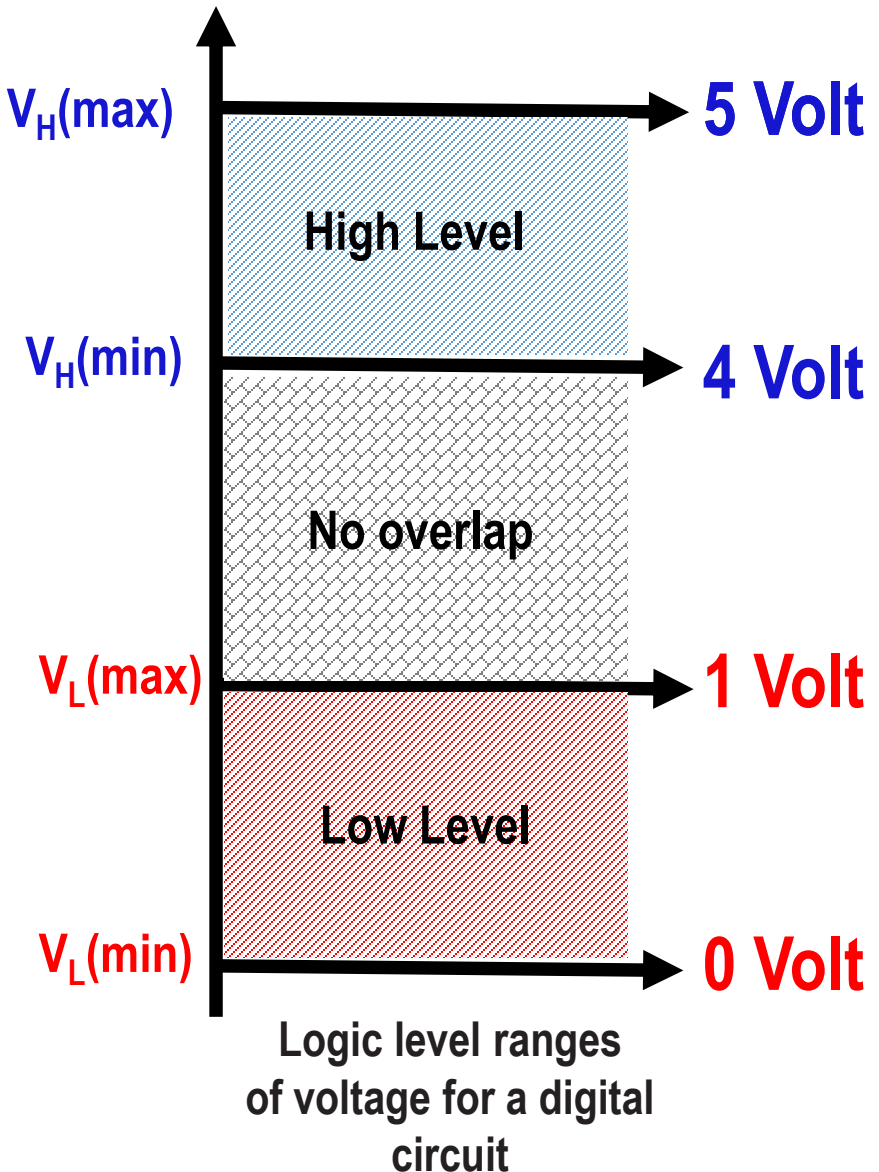


□ Logic Levels

□ The **voltages** used to represent a **High**→1 and a **Low**→0 are called **logic levels**.

□ In a **practical** digital circuit, a **HIGH** can be any voltage between a specified minimum value and a specified maximum value. Likewise, a **LOW** can be any voltage between a specified minimum and a specified maximum.

□ There can be **no overlap** between the accepted range of **HIGH** levels and the accepted range of **LOW** levels.

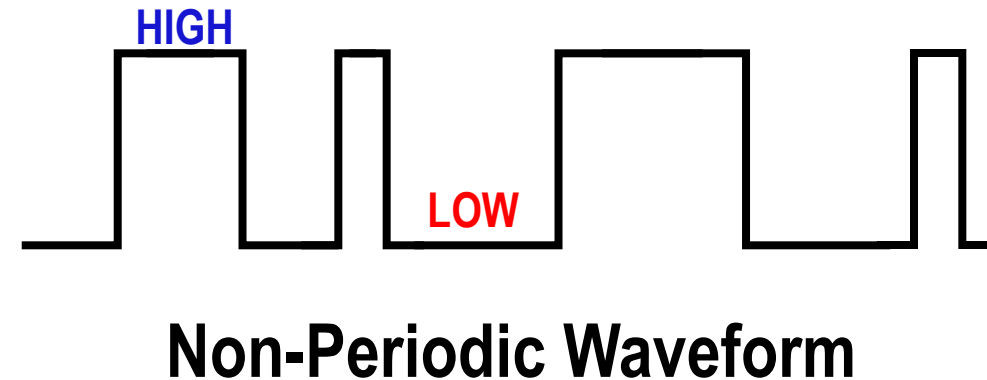
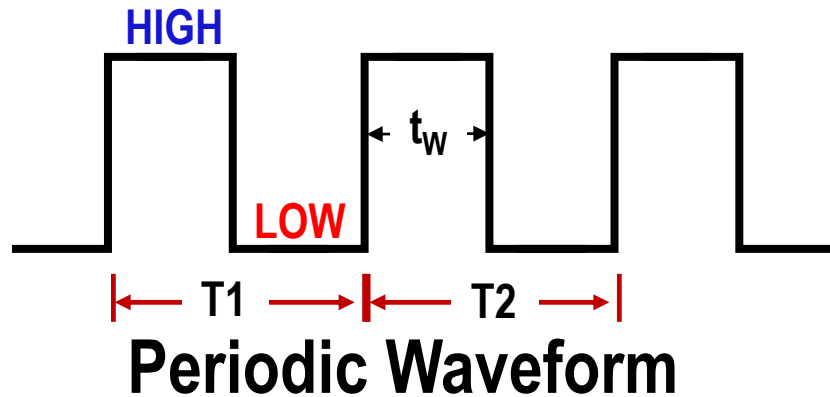


Binary Digits, Logic Levels, and Digital waveforms (3/5)



□ Digital Waveforms

- consist of voltage levels that are changing back and forth between the **HIGH** and **LOW** levels.
- are composed of **series of pulses**, sometimes called **pulse trains**



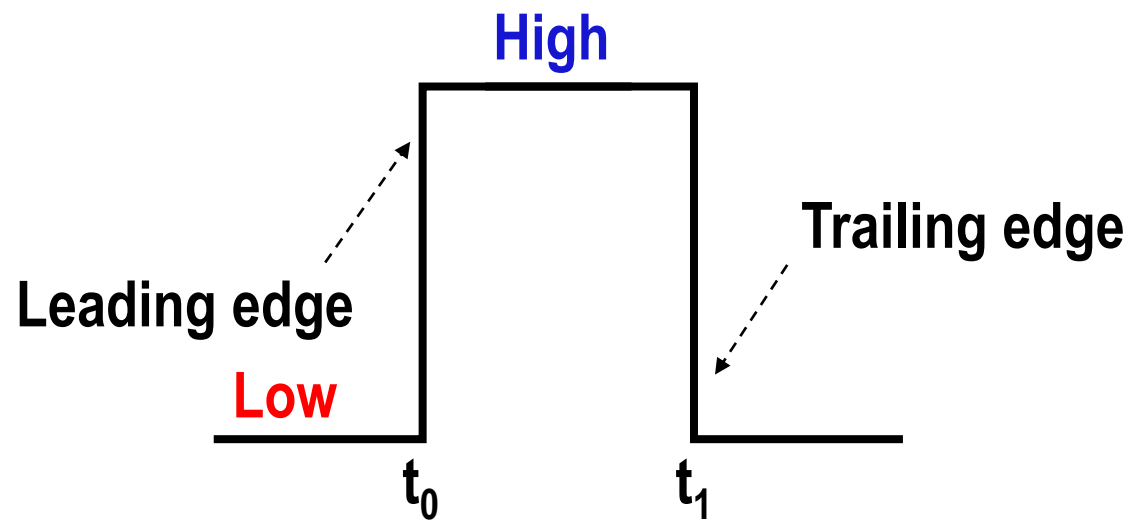
- Period = $T_1 = T_2 = \dots = T$
 - Frequency = $1/T$ (Hz)
 - An important characteristic of a periodic digital waveform is its **duty cycle**, which is the ratio of the **pulse width** (tW) to the **period** (T)
- ➡ Duty Cycle = $\left(\frac{T_W}{T}\right) * 100 \%$

Binary Digits, Logic Levels, and Digital waveforms (4/5)

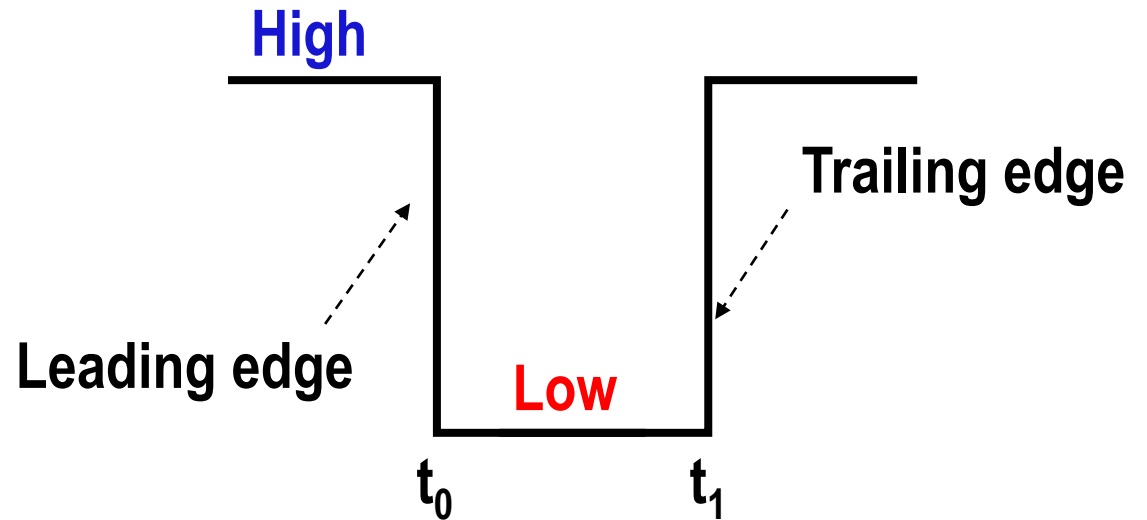


❑ Pulse's Characteristics

- ❑ A pulse has **two** edges: a **leading edge** that occurs first at time t_0 and a **trailing edge** that occurs last at time t_1



Positive-going pulse



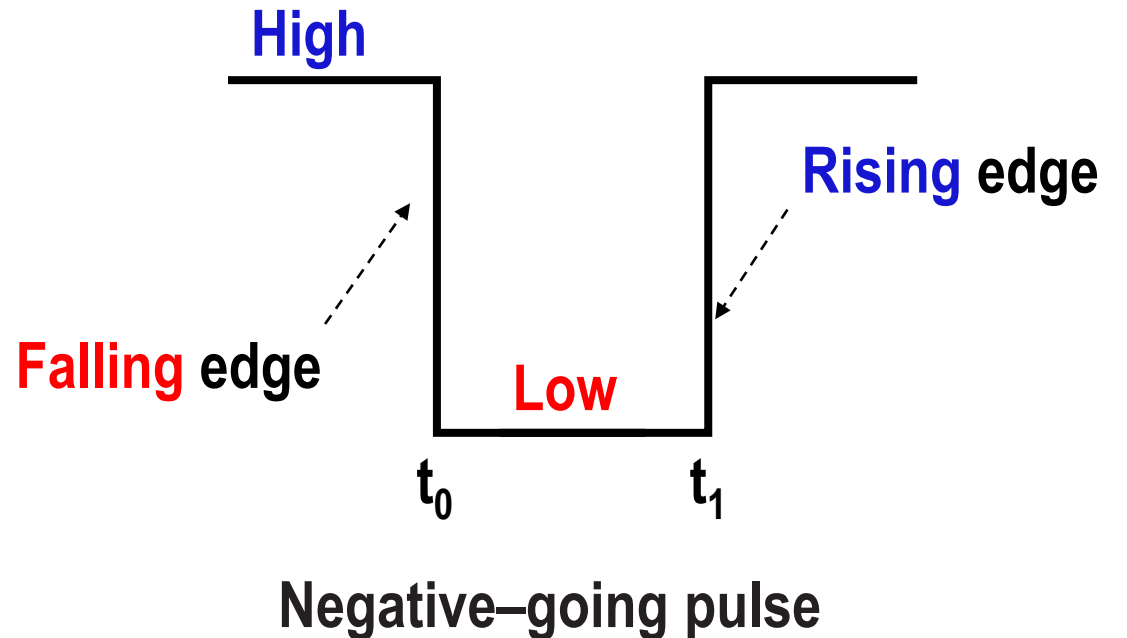
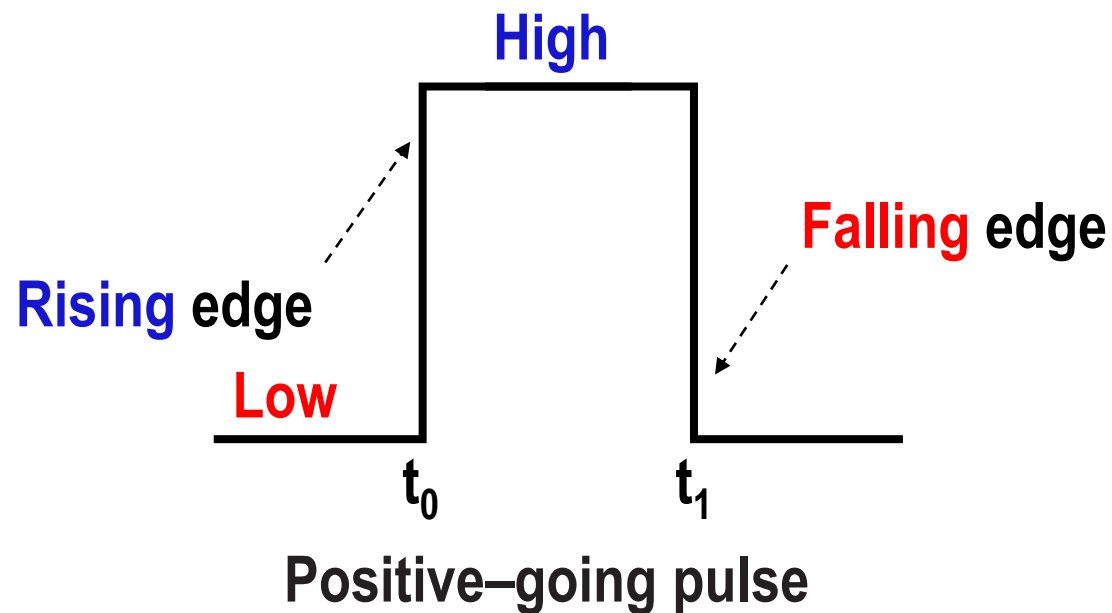
Negative-going pulse

Binary Digits, Logic Levels, and Digital waveforms (4/5)



❑ Pulse's Characteristics

- ❑ A pulse has **two** edges: a **leading edge** that occurs first at time t_0 and a **trailing edge** that occurs last at time t_1
- ❑ In ideal pulse, the **rising** and **falling** edges are assumed to change in **zero** time.

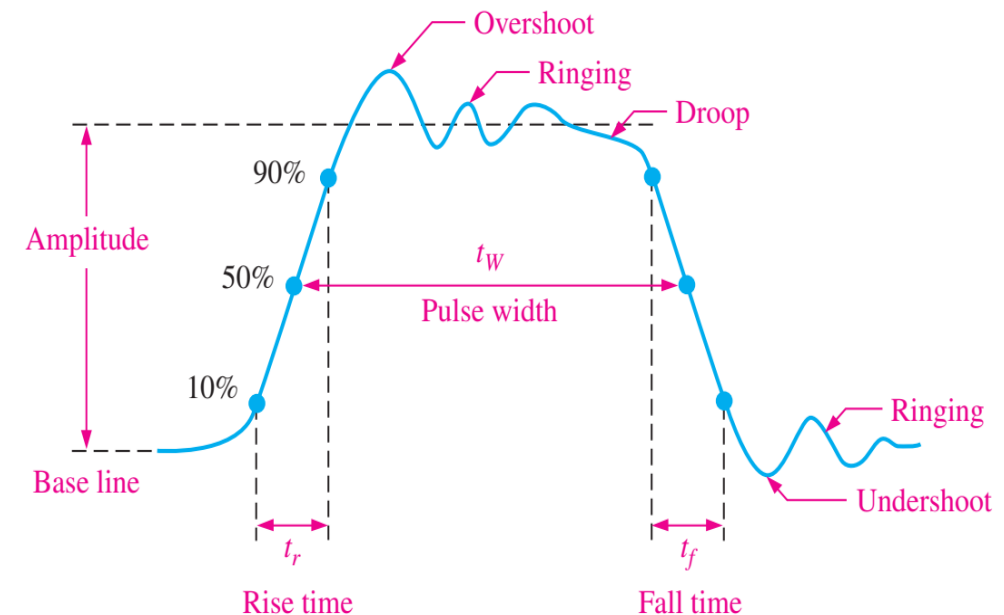
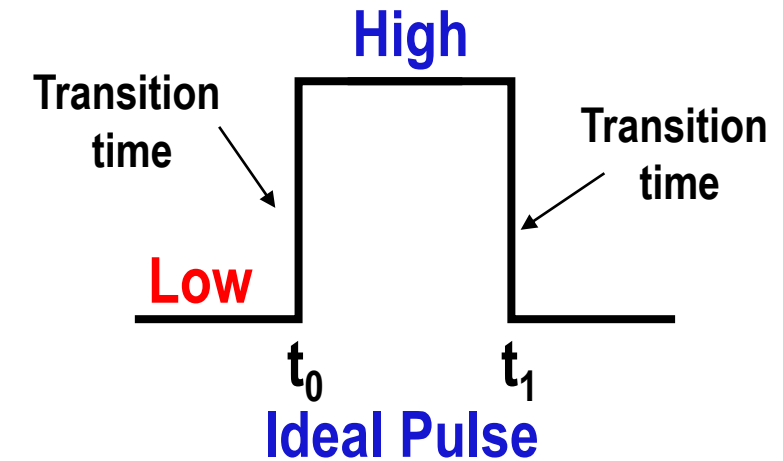


Binary Digits, Logic Levels, and Digital waveforms (5/5)



❑ Pulse's Characteristics

- ❑ In practice, these transitions **never** occur instantaneously.
- ❑ **Rise time** (t_r) from **10%** : **90%** of the pulse amplitude.
- ❑ **Fall time** (t_f) from **90%** : **10%** of the pulse amplitude.
- ❑ **Pulse width** (t_w) from **50%** points on the rising and falling edges.
- ❑ The **overshoot** and **ringing** are produced by stray **inductive** and **capacitance** elements.
- ❑ The droop can be caused by stray circuit **resistance**.



Non-Ideal (Practical) Pulse

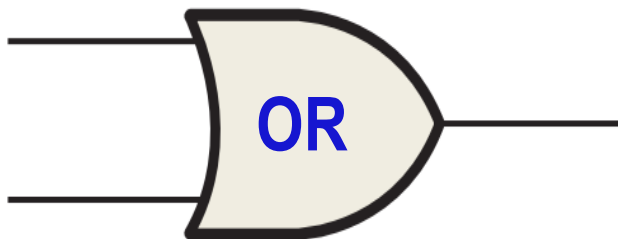
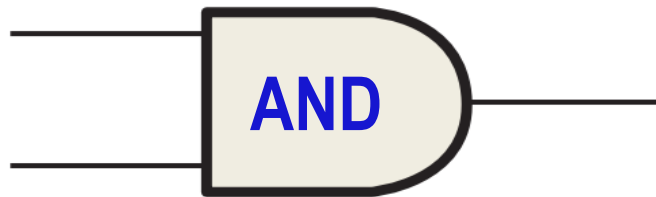
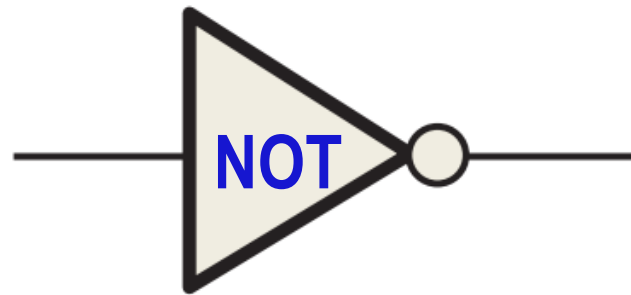


- ☐ Digital and Analog Quantities
- ☐ Binary Digits, Logic Levels, and Digital Waveforms
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- ☐ Programmable Logic
- ☐ Fixed-Function Logic Devices

Basic Logic Functions (1/2)



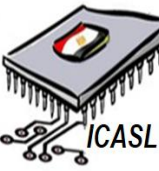
- ❑ The **three** basic logic functions AND, OR, and NOT



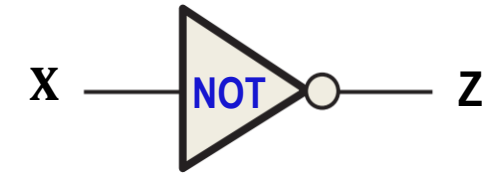
Form various other
types of more complex
logic functions

- ❑ Comparison
- ❑ Arithmetic
- ❑ Code conversion
- ❑ Encoding
- ❑ Decoding
- ❑ Data selection
- ❑ Counting
- ❑ Storage

Basic Logic Functions (2/2)

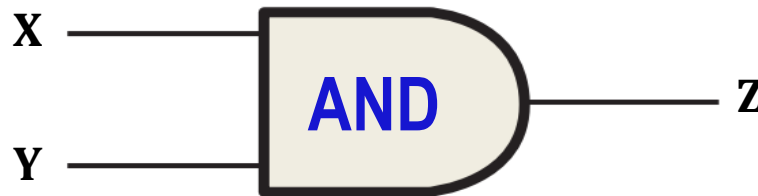


- ❑ The **NOT** function changes one logic level to the opposite logic level
- ❑ The **AND** function produces a **HIGH** output only when all the inputs are HIGH
- ❑ The **OR** function produces a **HIGH** output when one or more inputs are HIGH



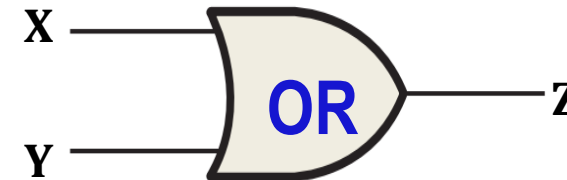
$$Z = \bar{X}$$

X	Z
0	1
1	0



$$Z = X.Y = XY$$

X	Y	Z
0	0	0
0	1	0
1	0	0
1	1	1



$$Z = X + Y$$

X	Y	Z
0	0	0
0	1	1
1	0	1
1	1	1

I/P O/P

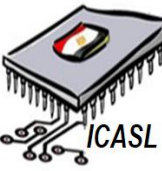
Note

- ❑ Truth table is a combinations of binary variables
- ❑ # of rows = 2^n , where n = # of input variables

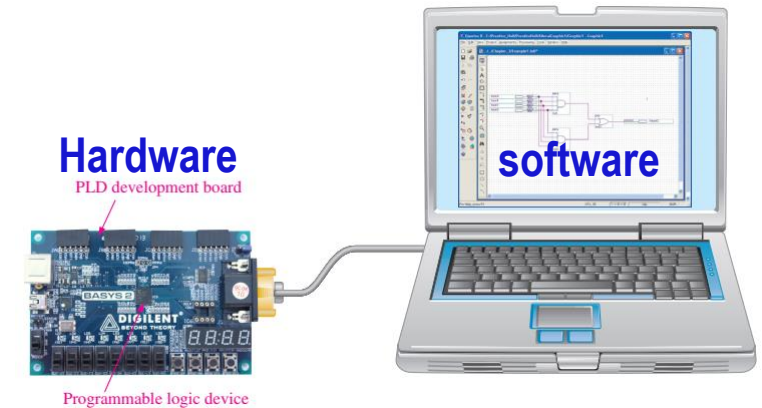


- ☐ Digital and Analog Quantities
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Programmable Logic (1/3)

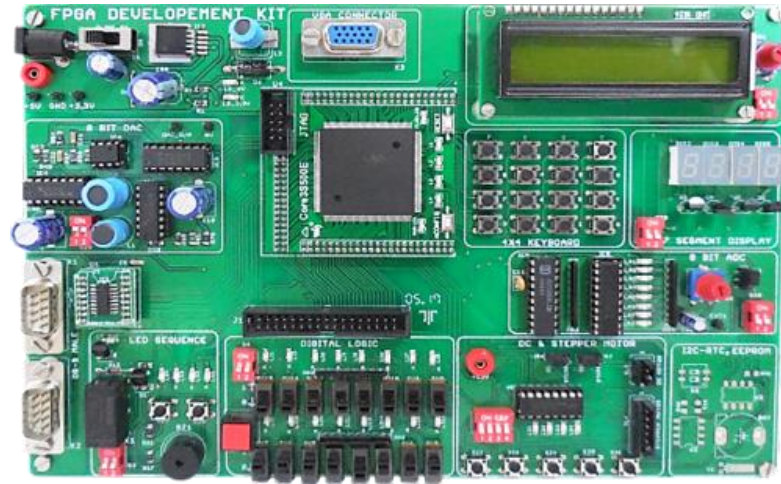


- ❑ Programmable logic requires both **hardware** and **software**.
- ❑ Programmable logic devices can be **programmed** to perform specified logic functions and operations by the **manufacturer** or by the **user**.

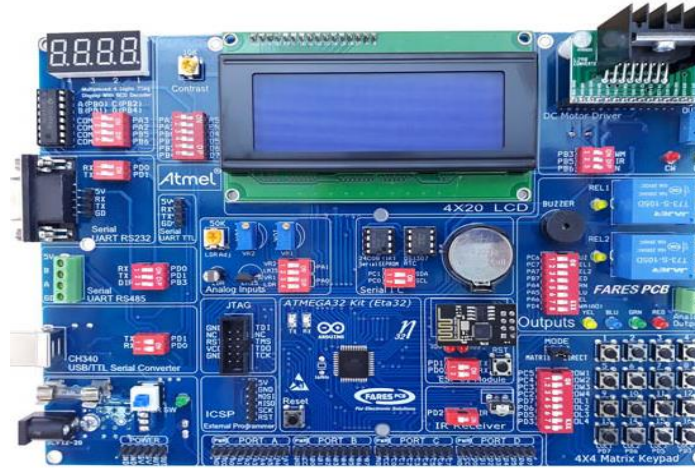


- ❑ **Advantages** of programmable logic.
 - ❑ Designs can be readily changed **without** rewiring or replacing components.
 - ❑ A logic design can generally be implemented **faster** and with **less cost**.
- ❑ **Disadvantage**
 - ❑ It requires **higher cost**.
 - ❑ It is **not efficient** to implement small segments of logic.

Programmable Logic (2/3)



Xilinx Spartan 3E FPGA Development Board



AVR Development Kits



Raspberry Pi RPI4-MODBP-4GB-BULK



Arduino Nano Arduino

VHDL
Verilog
Assembly
C,C++
Python

Programmable Logic (3/3)



User-programmable logic

PLD

(programmable logic device)

FPGA

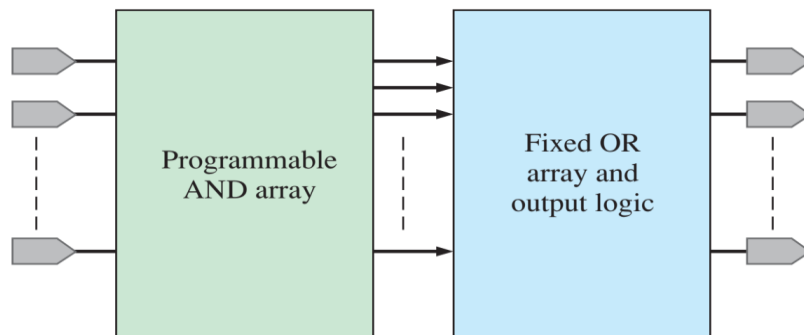
(field-programmable gate array)

Simple PLDs

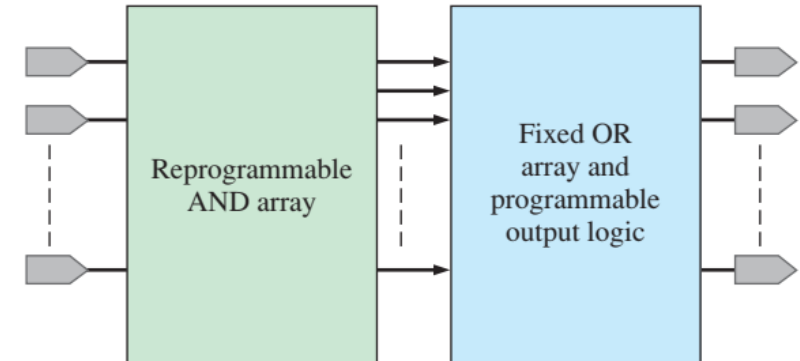
Complex PLDs

PAL (Programmable Array Logic)

GAL (Generic Array Logic)



Programmed **one time**



Reprogrammed **many times**

Programmable Logic (3/3)



User-programmable logic

PLD

(programmable logic device)

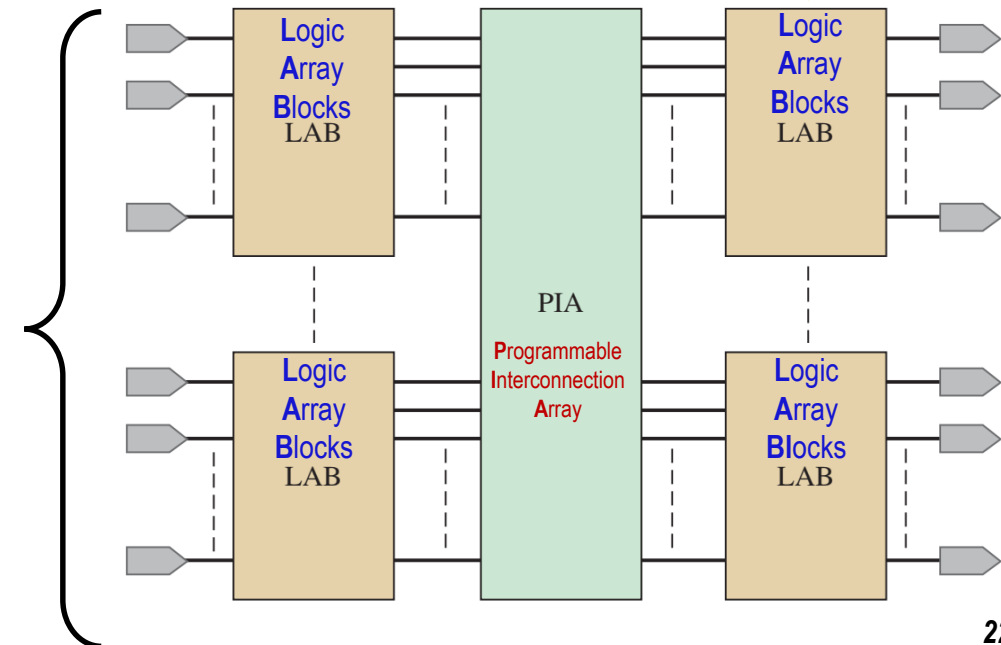
FPGA

(field-programmable gate array)

Simple PLDs

Complex PLDs

- ❑ Put more than one **SPLD** on a **single** chip
- ❑ Containing multiple **SPLDs** and can replace many fixed-function lcs.
- ❑ Implement any of the logic functions (decoders, encoders, multiplexers, demultiplexers, and adders)



Programmable Logic (3/3)



User-programmable logic

PLD

(programmable logic device)

FPGA

(field-programmable gate array)

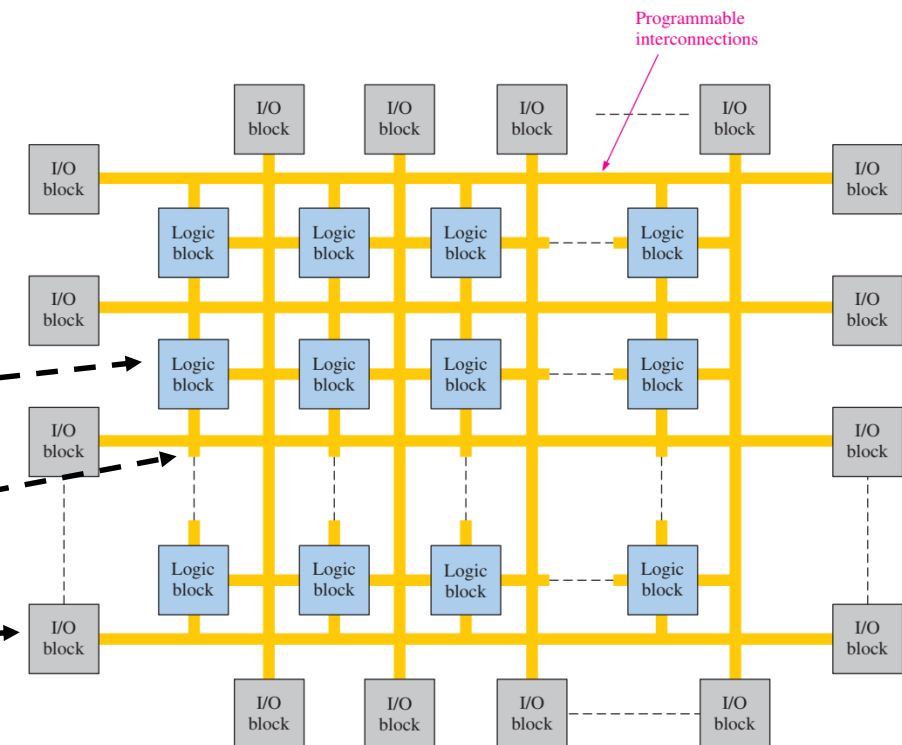
❑ An **FPGA** is generally more complex and has a much higher density than a **CPLD**

❑ **Three** basic elements in an FPGA are

❑ Logic block

❑ Programmable interconnections

❑ Input/output (I/O) blocks



Basic structure of an FPGA



- ☐ Digital and Analog Quantities
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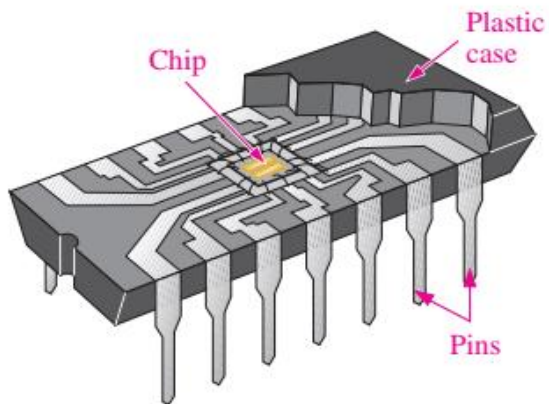
Fixed-Function Logic Devices (1/4)



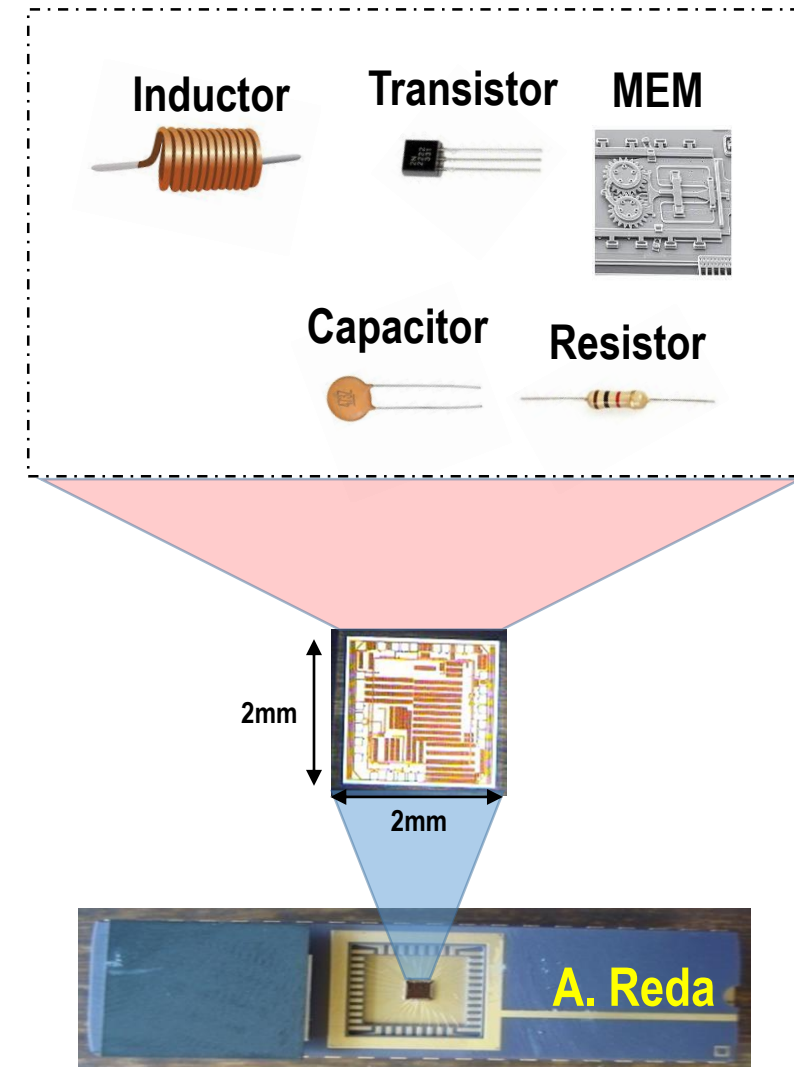
❑ An **integrated circuit** (IC) is a set of electronic circuits on one small flat piece (or "chip") of semiconductor material, usually silicon.

❑ Advantages of IC

- ❑ Small size
- ❑ High reliability
- ❑ Low cost for mass production
- ❑ Low power consumption



Cutaway view of one type of fixed-function IC package



Fixed-Function Logic Devices (2/4)

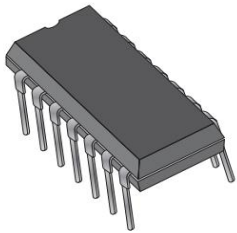


IC Package

Printed Circuit Boards (PCBs)

Chip-On-Board

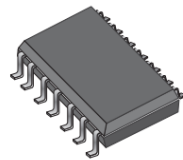
Through-hole mounted



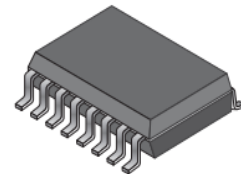
Dual in-line package (DIP)

Surface-mount Technology (SMT)

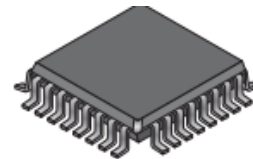
- The holes are unnecessary
- SM package is much smaller



Small-Outline
Integrated Circuit
(SOIC)



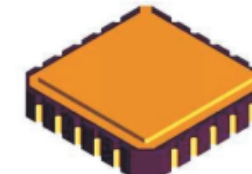
Shrink Small-
Outline Package
(SSOP)



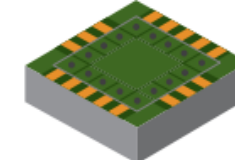
Low-profile Quad
Flat Package
(LQFP)



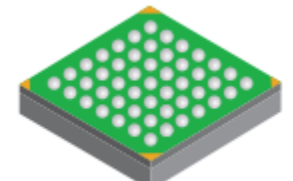
Plastic-Leaded
Chip Carrier
(PLCC)



Leadless
Ceramic Chip
(LCC)



Chip scale
package
(CSP)



Fine-Pitch Ball
Grid Array
(FBGA)

Gull-wing shape

J-type shape

Fixed-Function Logic Devices (2/4)



IC Package

Printed Circuit Boards (PCBs)

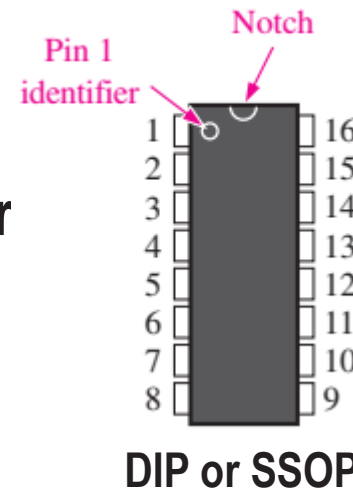
Chip-On-Board

Through-hole mounted

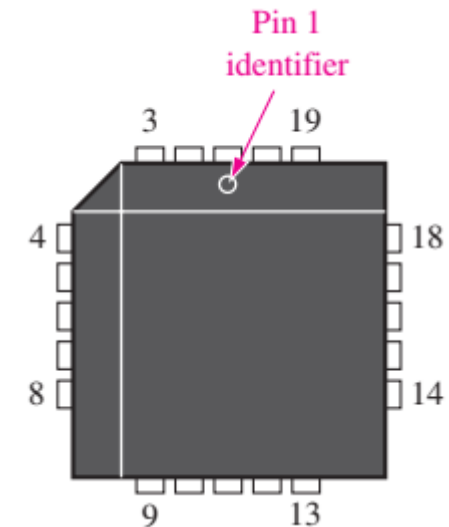
Surface-mount Technology (SMT)

Pin Numbering

- ❑ Looking at the top of the package,
- ❑ Pin 1 is indicated by a small dot, a notch, or beveled edge.
- ❑ The dot is always next to pin 1

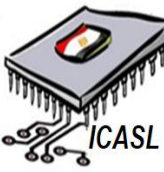


DIP or SSOP



DIP or SSOP

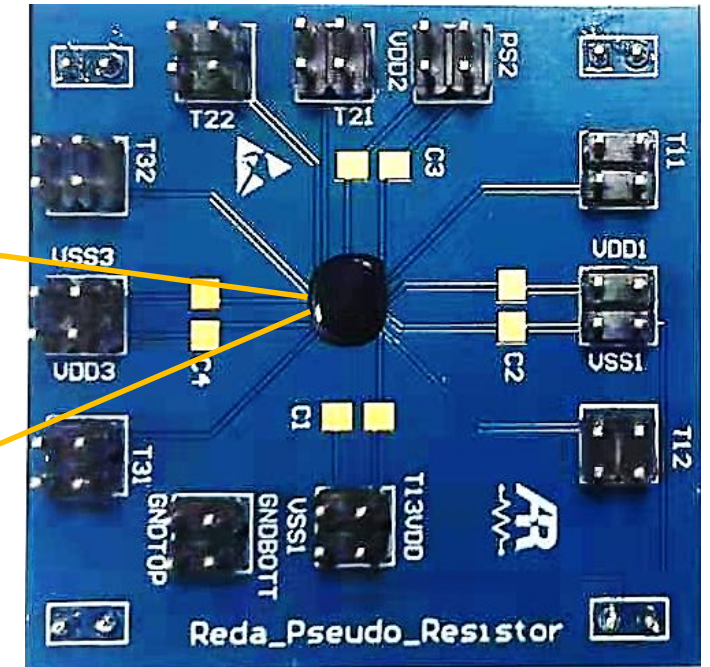
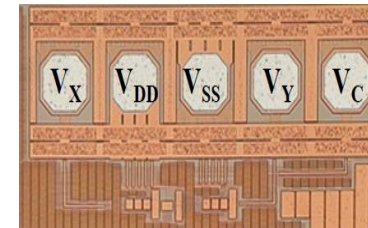
Fixed-Function Logic Devices (2/4)



IC Package

Printed Circuit Boards (PCBs)

Chip-On-Board

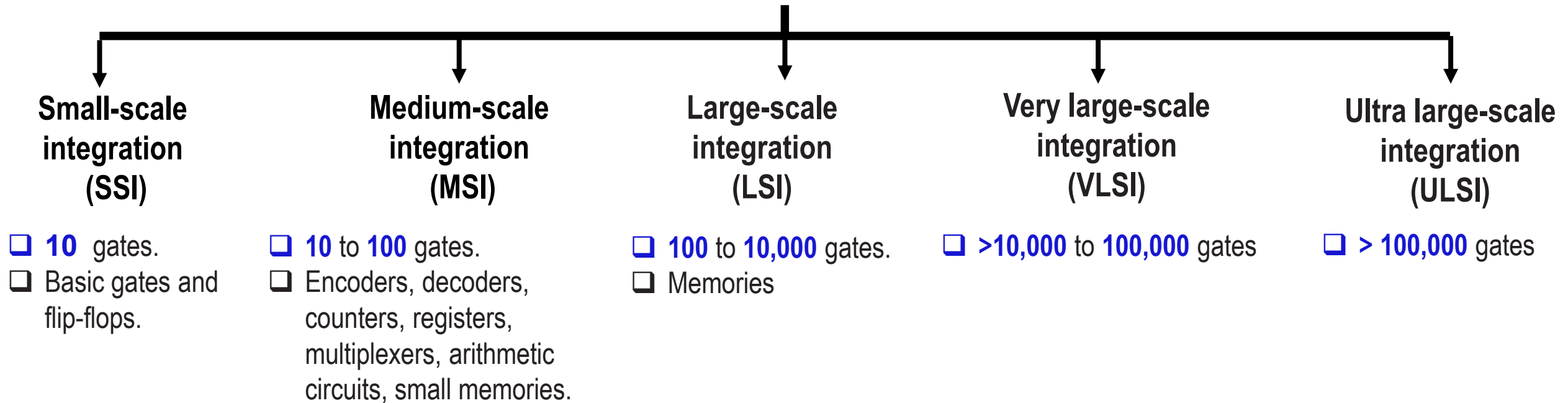


A chip-on-board (COB) is a chip that is mounted directly on a circuit board as opposed to being through-hole or surface-mounted technology.

Fixed-Function Logic Devices (3/4)



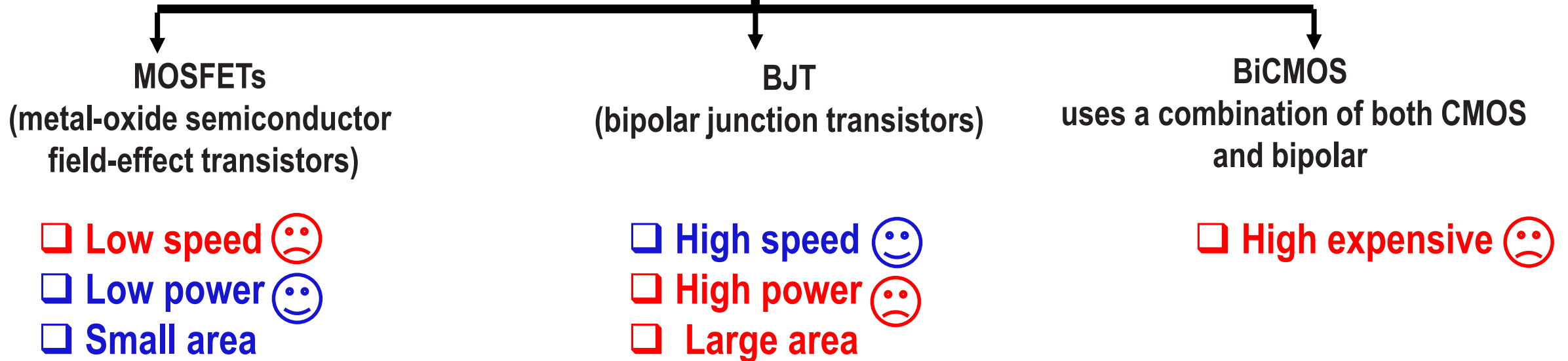
Complexity



Fixed-Function Logic Devices (4/4)



Technology



- ☐ **Advantage of CMOS technology** 😊
 - ☐ Cheapest technology for high density transistor.
 - ☐ Memory can be implemented density.
 - ☐ Fully restored level.
 - ☐ Very low static power.
 - ☐ Symmetrical device (symmetrical rise and fall time)

Thanks!